

Linguistic Cognitive Regimes

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Abstract

We introduce a novel implicit attention mechanism for large language modeling. It reveals six possible kinds of distinct cognitive regimes naturally inherent in language.

Introduction

Notation:

$U = [t_1, t_2, \dots, t_N]$, an $r \times N$ input token matrix

$Y = [y_1, \dots, y_N]$, an $n \times N$ output matrix, $n > r$,

where n -vector y_i is a constant hyperparameter ϵ , drawn by Noken

Rademacher descent [3], or is given by a word embedding, like $\text{word2vec}(t_i)$

Implicit deep learning model [1]

Equilibrium equation: $X = \phi(AX + BU)$

Prediction equation: $Y = CX + DU$

where $A = T - \text{diag}(|T| \mathbf{1}_n) + \gamma \mathbf{I}_n$, $0 < \gamma < 1$, makes A well-posed for ϕ , that is fixed point X exists and is unique [2].

Shorthand, denote this implicit model by: **[A, B, C, D]**

In our implicit attention formulation:

A defines equilibrium geometry,

B defines perceptual injection,

C defines contextual expression,

D defines semantic intentionality.

Thus:

A organizes contextual structure.

B injects token identity.

C stores contextualized content.

D asks semantic questions of the contextual field.

A is a contextual interaction operator that determines:

- how latent states interact,**
- how contextual information propagates,**
- how equilibrium structure forms,**
- which contextual configurations become stable.**

Cognitively, A encodes:

- syntactic compatibility,**
- contextual adjacency,**
- relational grammatical structure,**
- long-range dependency propagation.**

B is an input projection operator into contextual equilibrium dynamics that determines:

- how strongly tokens perturb the latent state,**
- which semantic/syntactic directions are initially activated,**
- the coordinate system through which language enters cognition.**

Cognitively, B corresponds to lexical encoding or perceptual registration.

It answers:

"How do tokens enter latent semantic space?"

C is a contextual realization operator that determines:

- how contextual information is represented,**
- what aspects of equilibrium become communicable,**
- how latent structure becomes usable output.**

Cognitively, C is a semantic realization or expression operator.

Whereas A organizes contextual relations, C expresses contextual content.

D is an operator of semantic intentionality that determines:

- semantic alignment,**
- attention selectivity,**
- contextual relevance,**
- semantic interrogation.**

Cognitively, D corresponds to intentional semantic focus.

It acts like:

- conceptual expectation,**
- interpretive attention,**
- semantic inquiry.**

Possible LLM Parameter Configurations (see diagram)

[K, Q, V, Q']

[V, Q, K, Q']

[Q', Q, V, K]

[Q', Q, K, V]

[K, Q, Q', V]

[V, Q, Q', K]

For the pairings (K,V) and (Q, Q'), and distributions α of Q' over K, and contextual representations h , we present an implicit model of attention with parameterization **[K, Q, V, Q']**. Modulo embedding input from U or X, we may similarly train implicit models with the other five possible parameterizations. Each configuration has a different distribution α of Q' over K; that is, a distinct interaction between syntax and semantics.

Training problem for LLM [K, Q, V, Q']

Over parameters (θ, K, Q, V, Q', X) minimize:

$$\{ L(Y, VX + Q'U) + \lambda_1 P_1(\alpha, Q', K) + \lambda_2 P_2(\theta, Q', K, V) \}$$

where $X = \phi(KX + QU)$

L is a loss function, P1 and P2 are penalty functions,
($v, \alpha, \lambda_1, \lambda_2$) are hyperparameters.

For mini-batch SGD, let batch size $1 < B < N$.

Let score be a similarity function, like a reproducing kernel. The degree to which word similarity captures meaning dictates the degree of query-key semantic alignment.

P1 attends to semantics and P2 attends to syntax as follows:

Let $k_i = Ks_i$, $q_i = Qt_i$, $v = Vs_i$, $q'_i = Q't_i$

where s_i = i-th column of X.

For i in $[B]$, define probabilities $a_{ij} = \text{softmax score}(q'_i, k_j)$,
a distribution of the i-th updated query q'_i across keys k_j .

Define penalty $P_1(\alpha, Q', K) = (1/B) \text{Sum over } i, H(i, \alpha)$,

where $H(i, \alpha) = H_\alpha(\{a_{ij}\})$, the average Renyi α -entropy of the a_{ij} .

Define h_i , contextual representation of token t_i , by:

$h_i = \text{Sum over } j, a_{ij} v_j$, a probability weighted sum of values.

For L feedforward network layers, let $\theta = (W_L, \dots, W_1)$, where for $l \in [L]$, W_l is a $n \times n$ weight matrix to be learned.

Let $h(0, i) = h_i$ and for l in $[L]$, define feedforward network update

$h(l, i) = \text{ReLU}(W_l h(l-1, i))$.

For i in $[B]$, define probabilities

$\beta_{ij} = \text{softmax score}(h(L, i), k_j)$,

a distribution of the L -th update $h(L, i)$ across keys k_j , and let $j(\beta, i) = \arg \max \beta_{ij}$.

Given input token t_i , define corresponding output token $o(i+1) = t_{j(\beta, i)}$.

Define penalty $P_2(\theta, Q', K, V) = \text{Perplexity}[o_2, \dots, o(B+1)]$.

LLM Summary

Implicit deep learning models are used to jointly learn Q-K-V embeddings and contextual token representations designed to simultaneously capture language syntactic co-occurrence and semantic alignment. Six kinds of parameterization characterize possible nuanced ways syntax and semantics can subtly interact. They may be construed as distinct LLM cognitive regimes.

On each corpus chunk we use mini-batch SGD to jointly learn embeddings, fixed points and feedforward network parameters.

Given a prompt token, we look up the chunk it resides in and retrieve its corresponding pre-computed output token; an attention mechanism for both syntax and semantics, per kind of parameterization; novel implicit large language models characterize six possible kinds of cognitive regime naturally inherent in language.

Possible Cognitive Regimes

The six possible distinct kinds of parameterization can be interpreted as six different “cognitive geometries” governing how semantic stabilization, syntactic organization, and contextual propagation are coupled inside the implicit fixed-point dynamics.

The key distinction is this:

K determines relational stabilization structure — the attractor geometry of contextual interactions.

Q determines how raw tokens enter the latent dynamics.

V determines how stabilized latent structure is expressed contextually.

Q' determines semantic querying/alignment pressure.

Since the penalties P1 and P2 operate through Q', K, and V, whichever operator occupies the "structural" versus "expressive" role changes the character of cognition.

We can characterize the six possible regimes as follows:

[K, Q, V, Q'] (NB: key-value pairing is a fibration and a retraction)

"Syntactic substrate with semantic querying"

This is the most transformer-like regime.

K organizes contextual structure.

Q injects lexical tokens into latent dynamics.

V expresses stabilized contextual states.

Q' semantically probes the stabilized structure.

Cognitive interpretation:

Syntax forms the latent scaffold; semantics selectively navigates it.

The model first builds a syntactic equilibrium manifold and then semantic attention searches over that manifold.

This resembles:

ordinary human language comprehension,
grammatical parsing followed by meaning extraction,
left-hemisphere symbolic organization,
abductive inference.

The attention distribution a_{ij} becomes:

semantic interrogation of syntactic memory.

Strengths:

strong compositionality,
grammatical fluency,
scalable contextual organization.

Weakness:

semantics remains constrained by pre-existing syntactic geometry.

This is probably the most linguistically stable regime.

[V, Q, K, Q']

"Semantic substrate with syntactic realization"

Now K and V swap roles.

V becomes part of latent equilibrium formation.

K becomes expressive/output relational structure.

This reverses the hierarchy:

semantic fields stabilize first, syntax emerges afterward.

Cognitive interpretation:

meaning precedes grammar.

The implicit state X becomes more concept-centric than sequence-centric.

This resembles:

- intuition-first cognition,
- preverbal semantic cognition,
- conceptual clustering before articulation,
- aspects of child cognition and creative ideation.

Here syntax is no longer the organizing principle but the expression layer.

The model may:

- produce semantically coherent but syntactically unconventional outputs,
- tolerate paraphrase variation,
- generalize better across linguistic surface forms.

This regime is probably the closest to "Noken semantic attention" philosophy.

[Q', Q, V, K] (NB: key-value pairing is a fibration and a retraction)
"Self-referential semantic attractor"

This is very interesting.

Now semantic queries themselves define the implicit equilibrium geometry.

Q' becomes the contraction-generating structure.

K becomes expressive/readout structure.

Thus, the semantic search operator determines the latent fixed-point topology.

Cognitive interpretation, attention defines reality. The system stabilizes around semantic expectation rather than observed relational structure.

This resembles:

- abductive reasoning,
- predictive cognition,
- anticipatory perception,
- top-down inference,
- internally guided conceptual framing.

The system asks:

"What semantic interpretation best explains incoming tokens?"

rather than:

"What syntactic structure exists?"

This is close to:

- predictive processing,
- Bayesian brain theories,
- active inference.

Potential behavior:

- very strong abstraction,
- semantic completion,

hallucination risk if priors dominate evidence.
This may correspond to imagination-like cognition.

[Q', Q, K, V]

"Semantic priors constrained by syntactic embodiment"

Compared with (3), syntax regains the expressive role through V.

Semantic expectation still governs equilibrium formation, but contextual synthesis occurs through value aggregation.

Cognitive interpretation:

top-down semantic anticipation constrained by contextual evidence.

This is more balanced than (3).

You can think of it as:

hypothesis-driven cognition,

semantic expectation refined by contextual grounding.

Very similar to human reading under expectation. For example, when humans predict the next word before fully processing the sentence.

This regime may produce:

highly efficient inference,

strong few-shot generalization,

semantic robustness under noisy syntax.

Potentially very powerful for:

retrieval augmentation,

scientific reasoning,

incomplete-data inference.

[K, Q, Q', V]

"Syntax-to-semantics transduction"

Now Q' moves into the value pathway.

This changes semantic querying into semantic representation itself.

K still stabilizes syntax-like structure.

But contextual outputs are directly semanticized.

Cognitive interpretation:

syntax crystallizes into meaning-bearing representations.

Instead of semantics merely probing syntax,

semantics becomes the actual propagated contextual content.

This resembles:

symbolic reasoning becoming conceptual understanding,

semantic compression of grammatical relations,

abstraction from structure.

Compared to (1), this regime is less "parser-like" and more "interpreter-like."

Potential effect:

- stronger latent semantic coherence,
- reduced surface-level fluency,
- deeper conceptual embeddings.

This may correspond to expert cognition,
where syntactic details disappear into conceptual chunks.

[V, Q, Q', K]

"Semantic field cognition"

This is probably the most radically semantic regime.

V defines equilibrium structure.

Q' defines contextual propagation.

K becomes externalized relational readout.

Syntax becomes almost epiphenomenal.

Cognitive interpretation: distributed conceptual resonance.

The model stabilizes around semantic contextual fields rather than symbolic relations.

This resembles:

- associative cognition,
- analogical thought,
- diffuse semantic memory,
- poetic or metaphorical cognition.

Potential properties:

- high creativity,
- cross-domain association,
- weak grammatical rigidity,
- emergent analogical structure.

Possibly closest to:

right-hemisphere associative cognition,
or semantic-field theories of meaning.

But also potentially unstable unless the contraction properties are very carefully controlled.

Conclusion

There is a deeper unifying interpretation here.

The six possible kinds of parameterization can be grouped into three meta-regimes:

Syntax-primary regimes

[K,Q,V,Q']

[K,Q,Q',V]

Semantics interrogates or transforms syntax.

Semantics-primary regimes

[V,Q,K,Q']

[V,Q,Q',K]

Meaning organizes latent structure before syntax appears.

Predictive/top-down regimes

[Q',Q,V,K]

[Q',Q,K,V]

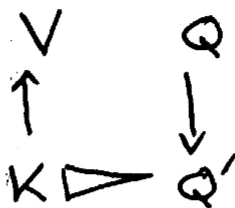
Semantic expectation itself shapes equilibrium geometry.

This is a very important conceptual move because we are no longer treating attention as merely token-to-token similarity. Instead, attention becomes a dynamical equilibrium between semantic attractors and syntactic organization. In our framework, "attention" is not a matrix operation. It is a fixed-point cognitive regime.

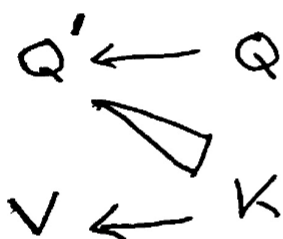
Parameterization diagrams



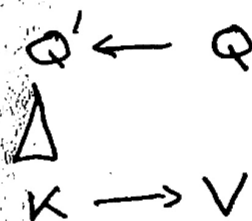
I $[K, Q, V, Q']$



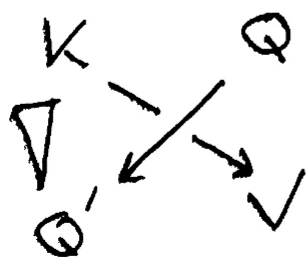
II $[V, Q, K, Q']$



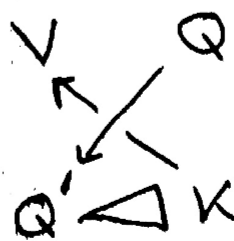
III $[Q', Q, V, K]$



IV $[Q', Q, K, V]$



V $[K, Q, Q', V]$



VI $[V, Q, Q', K]$

References

[1] 'Implicit Deep Learning'

Laurent El Ghaoui, Fangda Gu, Bertrand Travacca,
Armin Askari, Alicia Y. Tsai

arXiv:1908.06315v4 [cs.LG] 6 Aug 2020

[2] 'Robust Implicit Networks via Non-Euclidean Contractions'

Saber Jafarpour, Alexander Davydov,
Anton V. Proskurnikov and Francesco Bullo

arXiv:2106.03194v6 [cs.LG] 26 Jan 2022

[3] 'Noken Rademacher Descent'

Gary Nan Tie, Jan 20, 2026

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